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Toward rapid and efficient protocols: Synthesis of benzotriazole-based π-conjugated polymers via direct arylation polycondensation under aerobic conditions

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ABSTRACT

π-Conjugated polymers (CPs) are widely used in the field of high-performance plastic electronics. However, the development of sustainable and efficient methods for synthesizing donor-acceptor (D–A)-type CPs remains challenging for their industrialization. In this study, we introduce a simple and rapid direct arylation polycondensation (DArP) technique for preparing various D–A benzotriazole-based π-CPs with high molecular weights under aerobic conditions in two hours. Additionally, the method demonstrates not only excellent yields but also copolymers with nearly defect-free structures. Via a minor modification involving the use of twenty or more equivalents of the base K_2CO_3 and refluxing the solvent, high molecular weight copolymers can be achieved in a short reaction time under oxygen and moisture conditions. Therefore, this finding presents a direct synthetic protocol that utilizes commercially available and air-stable catalyst precursors to produce high-quality π-CPs suitable for optoelectronic applications.

1. Introduction

Over the last decade, numerous classes of organic semiconductor materials have been studied and attracted attention because of their great potential in plastic electronics [1–5]. Among organic semiconductor materials, π-conjugated polymers (CPs) have been applied successfully as active layers in optoelectronic devices due to their unique mechanical, physical, and chemical properties, including solution-processability and flexibility [6–10]. Various electronic devices such as organic light-emitting diodes, organic field-effect transistors, biosensor devices and polymer solar cells, have been fabricated and characterized for their performance properties [11–21]. As these technologies reach commercialization, the π-CPs should be required in larger quantities. Therefore, the availability of an efficient synthesis procedure for π-CPs is strictly considered. Typically, π-CPs have been synthesized based on traditional cross-coupling polycondensations such as Sonogashira, Heck, Kumada, Stille, Suzuki-Miyaura reactions, etc. [22–24]. However, these methods normally use organometallic reagents, releasing a stoichiometric amount of transition metal waste. They are usually toxic tin reagents requiring expensive treatment. Furthermore, conventional polycondensation methods involve several synthetic steps and necessitate the use of anhydrous solvents and oxygen-free environment. Consequently, additional procedures such as solvent distillation and storage under an inert atmosphere are required.

In recent years, direct arylation polycondensation (DArP) has been researched and applied as an efficient and sustainable method due to

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fewer synthetic steps, greater functional group tolerance and high-purity polymers under ideal conditions [25–31]. Moreover, the DArP method has recently attracted more attention compared with alternative traditional reactions because of the improvement of the molecular weight (M_n) of the resulting donor-acceptor (D-A) π-CPs [32–37]. However, most of the efficient Pd-catalyzed DArP systems are generally performed under oxygen and moisture-free conditions because of the sensitivity to oxygen and moisture of the Pd catalyst and intermediate compounds. Therefore, robust Pd catalysts have been designed to overcome catalytic deactivation in the presence of oxygen and moisture. Pioneered by Leclerc's group in 2017, the DArP in biphasic conditions was reported for the synthesis of isoindigo-based D-A polymers [38]. Lately, A. Ichige *et. al* described a facile synthesis of D-A π-CPs based on thienopyrroledione unit in the aerobic condition [39]. It should be noted that the polycondensation was performed in air by refluxing the solvents, without the need for additional catalysts or additives [39]. In 2020, Kanbara's group published a notable research for DArP. In this research, the XPhos Pd G2 was used as a precatalyst to synthesize π-CPs based on 3,4-ethylenedioxythiophene (EDOT) and 2,7-dibromo-9,9-dioctylfluorene (Br_2-DOF) in polar solvents to give a corresponding polymer with high M_n of 51 kDa in 24 h [40]. More recently, C. Carlucci *et. al* reported the synthesis of poly(9,9-dioctylfuorene-2,2′-bithiophene) using anisole solvent and the catalytic system composed of $(PPh_3)PdCl_2$/P(o-Anisyl)$_3$/Cs_2CO_3 under aerobic conditions [41]. The highest average M_n of the obtained polymers was about 11 kDa, which is relatively low compared to those obtained by other Still or Suzuki cross-coupling methods. However, most DArP in air has a long reaction time, usually more than 24 h to achieve high M_n polymers (M_n > 20 kDa), especially for D-A based on conjugated polymers based on polythiophene. Moreover, only a few studies with limited scope have successfully demonstrated the straightforward synthesis of D-A π-CPs. Notably, the DArP under oxygen and moisture conditions of low band-gap D-A terpolymers from three different monomer units remains a significant challenge.

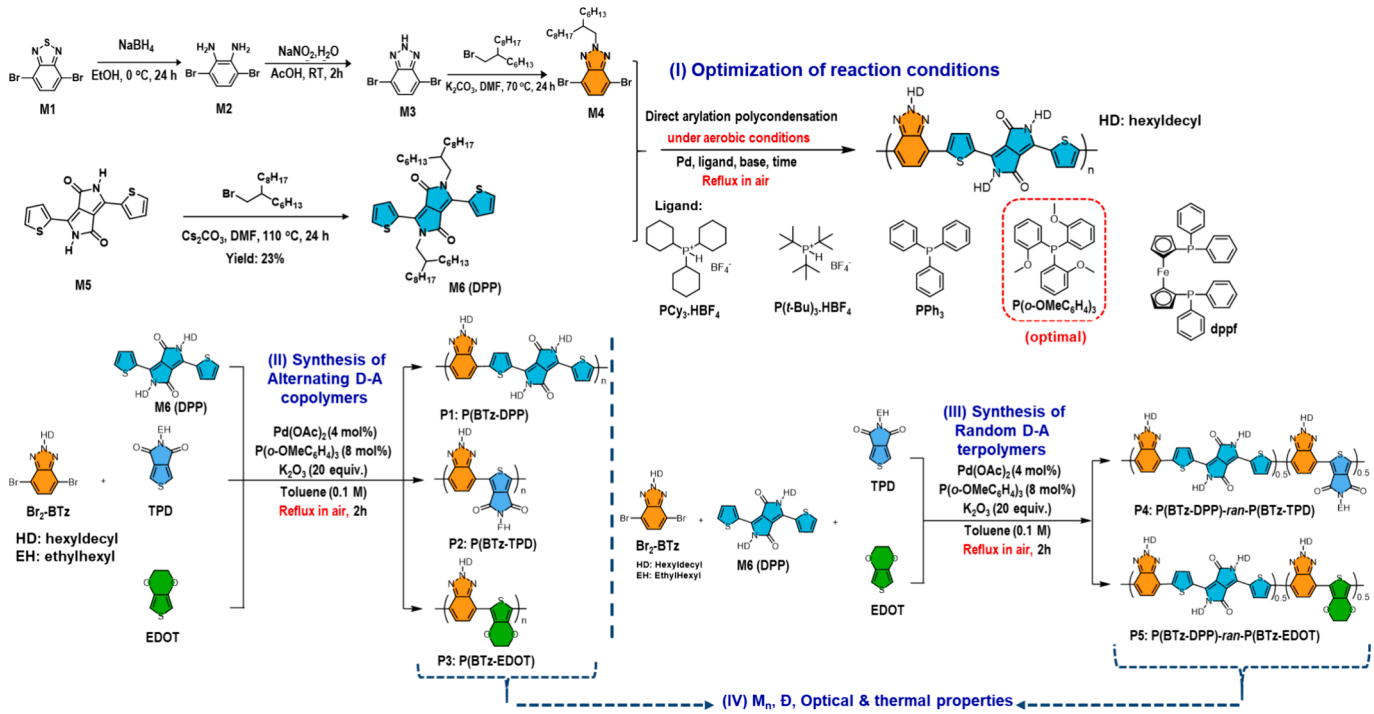
In this view, versatile catalyst systems and proper polymerization conditions as important parameters to give high M_n values of the resulting CPs and narrow polydispersity index (Đ) were considered. In this research, we investigated the aerobic DArP conditions to polymerize novel D-A CPs based on 2,5-bis(2-hexyldecyl)-3,6-di(thiophen-2-yl)pyrrolo[3,4-c]pyrrole-1,4(2H,5H)-dione and 4,7-dibromo-2-(2-hexyldecyl)-benzo[d][1,2,3]triazole where high polymer M_n could be achieved in a short reaction time. Furthermore, a variety of other new D-A-type copolymers and terpolymers were synthesized successfully with high M_n and narrow Đ by a single DArP reaction under aerobic conditions.

2. Results and discussion

Scheme 1 shows the scheme representing the flowchart of the experimental procedure to synthesize BTz-based donor-acceptor copolymers under aerobic conditions.

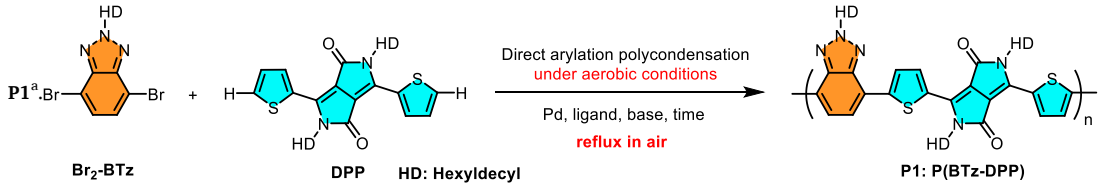
Aerobic DArP studies of P(BTz-DPP). We selected the polycondensation of dibromide benzotriazole (**Br_2-BTz**) and diketopyrrolopyrrole (**DPP**) as a model reaction to investigate aerobic DArP towards D−A π-CPs. The purpose was to study the effect of reaction parameters on the M_n and Đ of the polymer. The low-bandgap alternating π-CP **P (BTz-DPP)** has attracted attention for its potential in plastic electronic devices [42,43]. In this research, we used commercially available reagents without further purification to design a straightforward and potentially scalable protocol. All reactions were conducted by refluxing the solvent under air. The concentration of the monomer solution was set to 0.1 M to prevent the reactor from gelation, which can occur due to increases in solution viscosity as the reaction progresses and the molecular weight increases. After the reactions, the polymers were precipitated in methanol, filtered and dried in the oven overnight. They were then used for analysis without undergoing further Soxhlet purification. For detailed results, please refer to the Supporting Information (Table S1).

Based on the previously reported catalyst systems by Kanbara and coworkers for DArP under aerobic conditions [39], we initially synthesized P1 using $Pd(OAc)_2$, $PCy_3.HBF_4$, Cs_2CO_3, PivOH, and reagent-grade toluene at reflux for 24 h under air. Although the polymer was successfully produced, it had a moderate M_n of 7.5 kDa (Table 1, entry 1). Since the ligand plays a crucial role in achieving high molecular masses for polycondensation, we tested various phosphine ligands.



Scheme 1. Scheme representing the flowchart of the experimental procedure to synthesize benzotriazole (BTz)-based copolymers.

Table 1
Optimization of DArP under aerobic conditions for



Entry	Catalyst	Ligand	Base (equiv.)	Solvent (v/v)	Yield (%) [b]	M_n (kDa) [c]	Đ [c]
1	$Pd(OAc)_2$	$PCy_3.HBF_4$	Cs_2CO_3 (3)	Toluene	70	7.5	2.32
2	$Pd(OAc)_2$	$P(o\text{-}OMeC_6H_4)_3$	Cs_2CO_3 (3)	Toluene	87	17.8	2.15
3	$Pd(OAc)_2$	$P(o\text{-}OMeC_6H_4)_3$	K_2CO_3 (3)	Toluene	90	18.2	2.01
4	$Pd(OAc)_2$	$P(o\text{-}OMeC_6H_4)_3$	K_2CO_3 (20)	Toluene	94	39.2	1.84
5	$Pd(OAc)_2$	$P(o\text{-}OMeC_6H_4)_3$	K_2CO_3 (40)	Toluene	93	39.6	1.88
6 [d]	$Pd(OAc)_2$	$P(o\text{-}OMeC_6H_4)_3$	K_2CO_3 (20)	Toluene/H_2O (10/1)	89	25.5	2.03
7 [d]	$Pd(OAc)_2$	$P(o\text{-}OMeC_6H_4)_3$	K_2CO_3 (20)	Toluene/H_2O (1/1)	86	18.6	2.54

[a] Reaction conditions: **Br_2-BTz** (0.1 mmol, 1.0 equiv.), **DPP** (0.1 mmol, 1.0 equiv.), Pd catalyst (4 mol %), ligand (8 mol %), PivOH (1.0 equiv.), base, pristine reagent-grade solvent (2.0 mL), under reflux in the air for 24 h. [b] Isolated yield of the product after collecting the precipitated polymer. [c] M_n: number-averaged molecular mass, Determined by GPC calculated with polystyrene standards at 40 °C in chloroform. [d] Used tetrabutylammonium bromide as a phase transfer agent (1 equiv.).

Surprisingly, $P(o\text{-}OMeC_6H_4)_3$ showed highly effective polycondensation and resulted in a much higher M_n (17.8 kDa) in 87 % yield (Table 1, entry 2), attributed to its strong catalytic activity [38,44,45]. On the other hand, other ligands, such as $P(t\text{-}Bu)_3.HBF_4$, PPh_3, and dppf led to polymers with lower M_n (5.8 kDa, 3.4 kDa, and 5.3 kDa, respectively) but undesired high Đ (Table S1, entries 2–5). In sharp contrast, the absence of a ligand produced only a small amount of oligomer (Table S1, entry 6). These observations indicate that steric hindrance $P(o\text{-}OMeC_6H_4)_3$ is beneficial in improving M_n and yields for polycondensation under aerobic conditions.

To improve reactivity, we examined various reaction conditions. Interestingly, when we substituted Cs_2CO_3 with K_2CO_3, alternating copolymer **P1** with a slightly higher molecular mass (M_n = 18.2 kDa, Đ = 1.81) in 90 % yield was obtained (Table 1, entry 3). It is not easy to claim that K_2CO_3 is superior to Cs_2CO_3, but the truth is that the choice of K_2CO_3 arises from its wide availability and inexpensiveness [38]. To enhance the M_n and yield of **P1** under aerobic conditions, we also explored different palladium precursors (i.e., $Pd_2(dba)_3$, $PdCl_2$, and $PdCl_2(PPh_3)_2$) without observing any improvement in the reaction (Table S1, entries 8–10). The best results, in terms of molecular weights and yields, were achieved with $Pd(OAc)_2$. Indeed, $Pd(OAc)_2$ is widely known as an effective catalyst that performs admirably in many cross-coupling reactions under aerobic conditions [46,47].

The amount of base used was found to affect the reaction rates. Increasing the amount of base can potentially enhance the transmetalation rate, which in turn facilitates polycondensation and prevents side reactions [26,38]. To test this, we increased the amount of K_2CO_3 base to 10 equivalents, resulting in a significant increase in both molar mass (M_n = 36.5 kDa) and yield (92 %) (Table S1, entry 11). Further increasing the base to 20 and 40 equivalents led to even higher molecular weights of 39.2 kDa, and 39.6 kDa, with yields of 94 % and 93 %, respectively (Table S1, entries 12–13). As expected, the reaction did not occur without a base (Table S1, entry 14). Studies on the impact of K_2CO_3 stoichiometry verified that the ideal K_2CO_3 concentration was 20 equivalents. These results highlight the importance of using a base for DArP. Additionally, we conducted the polycondensation under N_2 using the freeze–pump–thaw treatment and obtained a polymer with a similar molar mass (M_n = 38.5 kDa, Table S1, entry 17) to that achieved under air. This demonstrates that our DArP methodology is a promising approach for accessing a wide range of high molecular mass alternating D−A π-CPs, without the need for inert protection and anhydrous conditions.

Polycondensation kinetics. During the optimization studies conducted to develop more robust DArP in the presence of moisture and oxygen, we observed distinct differences in polycondensation behaviours under various conditions. We therefore monitored the reaction process over time and studied the impact of reaction parameters on the M_n and Đ of copolymer **P1** (Table 1 and Fig. 1). Firstly, we noticed a clear difference in polycondensation rates when comparing the effects of two ligands, $PCy_3.HBF_4$ and $P(o\text{-}OMeC_6H_4)_3$. When $P(o\text{-}OMeC_6H_4)_3$ was used as a ligand, M_n of **P1** was significantly higher, roughly twice as high as when $PCy_3.HBF_4$ was used (Table 1, entries 1–2, and Fig. S3-4). This finding aligns with previous studies that have reported the enhanced efficiency of $P(o\text{-}OMeC_6H_4)_3$ on the reaction rate [28,45,48]. The presence of the base Cs_2CO_3 or K_2CO_3 resulted in similar reaction rates. Notably, using K_2CO_3 as a base led to an increase in both M_n and yield, reaching values of 18.2 kDa and 90 %, respectively (Table 1, entry 3, and Figure S5). This result confirms the efficacy of K_2CO_3 as a base, which is consistent with previous reports [49,50].

Increasing the equivalent of K_2CO_3 from 3 to 20 significantly accelerated the reaction rate, resulting in an M_n of **P1** reaching nearly 40 kDa within 2 h (see Table 1, entries 4–5, Table S1 and Figure S6-7). The use of electron-poor and bulky phosphine $P(o\text{-}OMeC_6H_4)_3$ tends to increase the rate of the OA step, and reduce the formation of homocoupling and β-arylation defects [32,45]. Moreover, increasing the content of K_2CO_3 facilitates the formation of $PivO^-$ ions, affecting the formation of Bidentate 1 and Concerted Metalation Deprotonation (CMD) [50]. CMD is an intermediate compound that plays a critical role in the deprotonation of the thiophene ring (the most crucial step in DArP), thus enhancing the polymerization rate [51]. Additionally, refluxing toluene

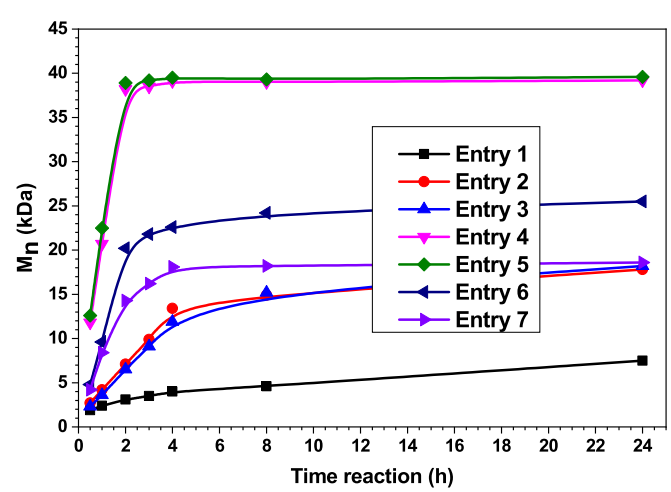


Fig. 1. The relationship between M_n and the reaction time of **P1** under different conditions in Table 1.

solvent is essential for the successful execution of the DArP reaction under aerobic conditions.

As previously mentioned, the impact of adding water to DArP remains a subject of controversy [38–41,49,52]. In our study, we examined how the volume ratio of water to toluene affects the arylation reaction. The results, presented in Table 1 (entries 6–7) and Figure S8-9, showed that increasing the volume ratio of water led to a significant reduction in the M_n of polymer **P1**, even when a large amount of K_2CO_3 and a long reaction time was applied. These results suggest that the conclusions drawn by the Lercler group may only apply to a specific polymer system [38]. These findings suggest that a reasonable water content in toluene might induce the dissolution of the base (K_2CO_3) leading to a faster reaction rate, thereby enhancing the molecular weight. Conversely, excessive water could limit the molecular weight because it would cause an early precipitation of the conjugated polymer. In biphasic conditions, water and toluene separate into two phases. With a higher water content, the diameter of the water bubbles tends to increase during the reaction process. This can reduce the contact area between the water phase (containing the base) and the toluene phase (where the polymer synthesis reaction takes place), thereby decreasing the efficiency and M_n of the polymer [27].

Polycondensation scope. Encouraged by the success mentioned above, we next examined the applicability of the present aerobic DArP for other π-CPs (Fig. 2). Among the efficient monomers for reactivity towards C−H functionalization, N-alkylthienopyrrolodione (TPD) is a logical choice for producing various D−A-type important π-CPs for high-performance organic electronics [53,54]. The copolymerization of Br_2-BTz and TPD, as two electron-deficient monomers, afforded an alternating BTz-TPD polymer (**P2**) with a good M_n value (31.8 kDa) and excellent yield (95 %) in a short reaction time (2 h) under optimal conditions. Furthermore, we attempted to expand the scope of the aerobic DArP by using Br_2-BTz with an electron-rich monomer 3,4-ethylenedioxythiophene (EDOT), as EDOT is known to enhance the quinoidal character of π-CPs, allowing them to absorb long-wavelength visible light and improve charge transfer ability [55]. However, the synthesis of P(BTz-EDOT) (**P3**) under the same condition led to a decrease in the M_n (M_n = 24.5 kDa), presumably due to the unsuitable effect of toluene on the electron-rich thiophene EDOT [56]. Nonetheless, compared with previous reports [57,58], this result still shows outstanding advantages in terms of high molecular weight, excellent yield and short reaction time.

After successfully preparing various alternating D−A π-CPs, we expanded this aerobic DArP method to synthesize conjugated random terpolymers (Fig. 2b). These terpolymers combine three components as repeating units into a backbone. Terpolymers are promising macromolecules for high-performance organic electronics due to their perfectly defined complementary absorption, optimized molecular energy levels, improved blend morphology, and enhanced solubility and processability [15,59]. We investigated the random terpolymerization of three electron-deficient units (1 equiv. of Br_2-BTz with 0.5 equiv. of DPP and 0.5 equiv. of TPD) to obtain A–A–A-type π-CPs (**P4**). Impressively, this polymer exhibited a high M_n value (37.3 kDa) and excellent yield (91 %). Moreover, we successfully produced a high-molecular-weight A–A–D-type terpolymer **P5** (M_n = 29.1 kDa) under identical conditions. This terpolymer consists of EDOT as a donor and two acceptor units (BTz and DPP) and demonstrates the general applicability of our protocol.

Polymer structure. The chemical structures of the obtained polymers **P1-P5** were characterized by ^{1}H NMR to evaluate the structural defects (Fig. 3). The main chemical shifts of P1 were essentially in agreement with those of **P1** obtained from other polycondensations in an oxygen-free environment (e.g., DArP and Suzuki–Miyaura catalyst-transfer polymerization [57]), and did not display any observable resonances corresponding to homocoupling units. The integral ratios of the signals (A_1 and C_1) agreed with the assignments and were consistent with the strictly alternating structure BTz-DPP in **P1** (Figure S13). However, analysis of the terminal groups of this polymer was not possible because of the broadening of the ^{1}H NMR signals. The ^{1}H NMR spectrum of **P2** not only showed clearly that the main signals (A_2 and C_2) were consistent with the repeating BTz and TPD units, but also homocoupling defects were undetected in the polymer backbone (Figure S14). The ^{1}H NMR spectrum of **P3** also indicated that no homocoupling defects were present and an alternating D–A structure of EDOT and BTz was the dominant structure (Figure S15), which was consistent with previous reports [57,58]. Similarly, the composition and participation ratio of the components in the random terpolymers were consistent with

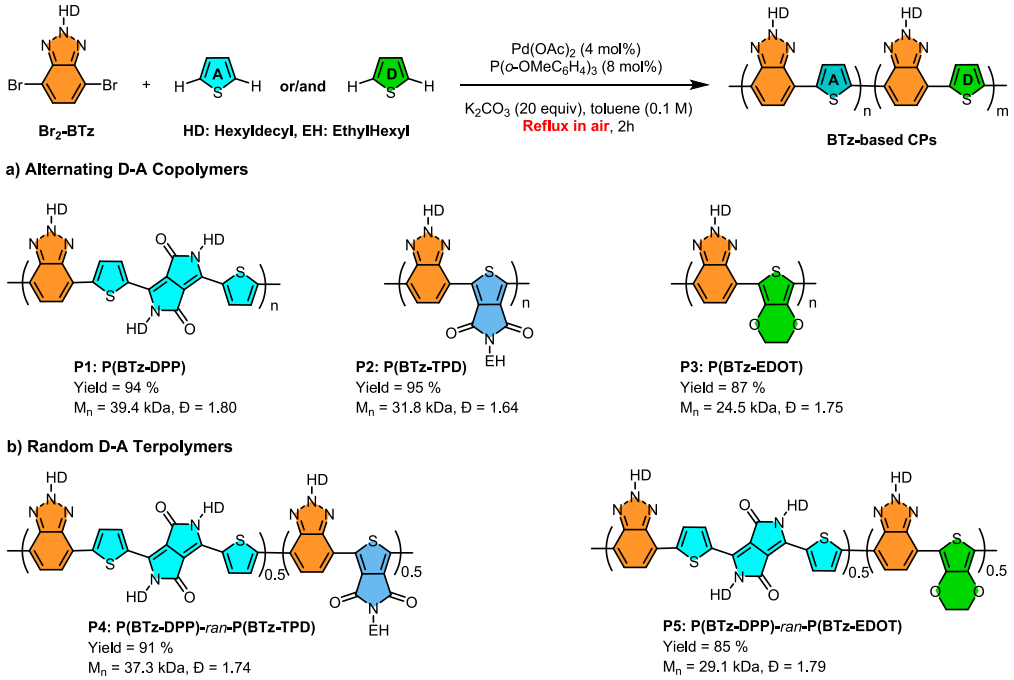


Fig. 2. Synthesis of BTz-based π-CPs using aerobic DArP optimization.

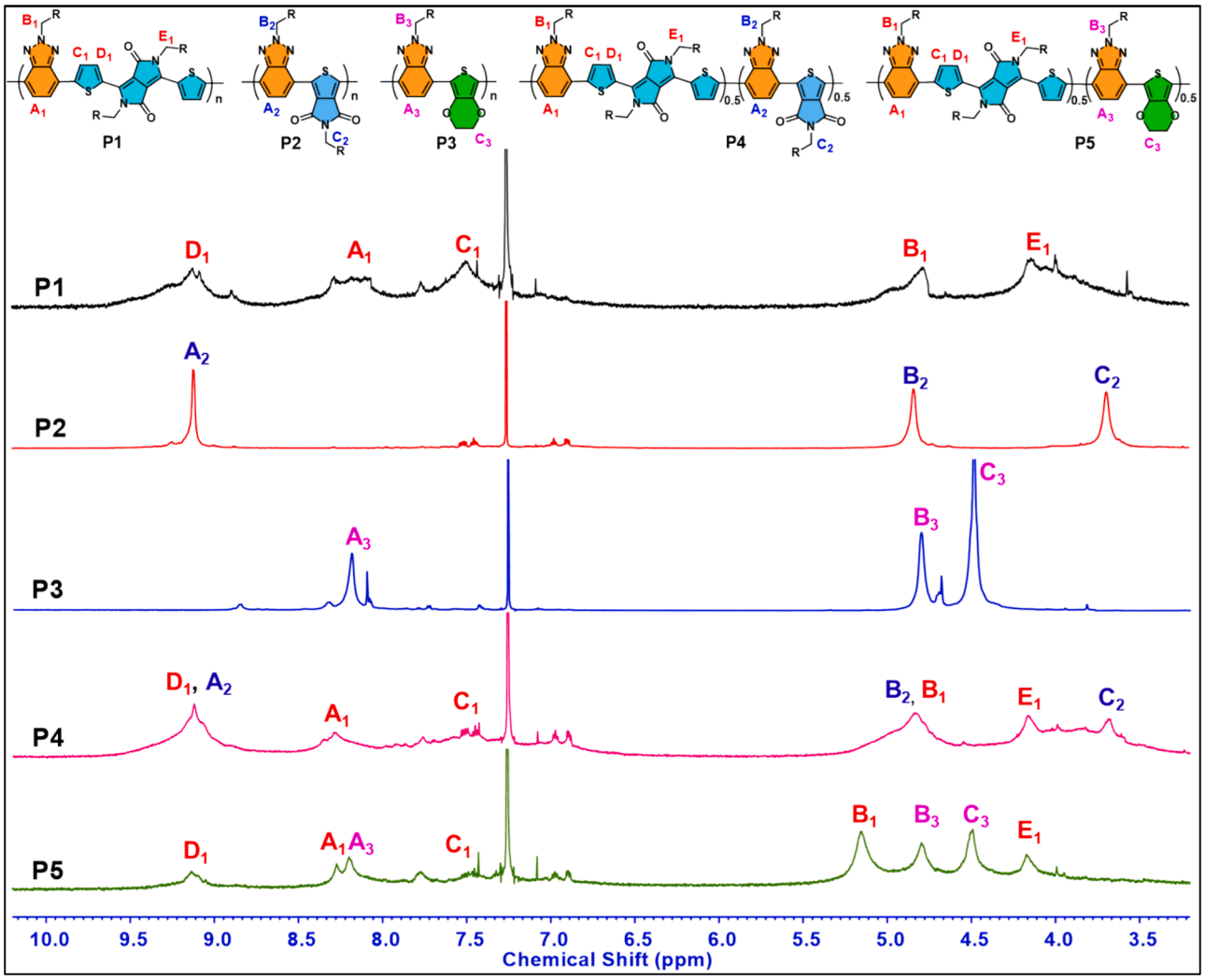


Fig. 3. ^{1}H NMR spectra of **P1-P5** ($CDCl_3$, 500 MHz, 298 K).

the structural design. Furthermore, peaks corresponding to homocoupling and β-arylation defects were undetected in the ^{1}H NMR spectrum of **P4** and **P5** (Figure S16 and Figure S17). The chemical structures of the obtained polymers **P1-P5** were also confirmed by FTIR analysis (Figure S18). These results indicate that our DArP protocol is advantageous for achieving both alternating D–A copolymers and random terpolymers with extremely low defect levels as well as high M_n.

Photophysical properties. The UV–Visible (UV–Vis) absorption measurements of the BTz-based π-CPs **P1-P5** were conducted to investigate the impact of structural defects on the optical properties of these materials. The measurements were performed in a dilute chloroform solution (1.0×10^{-5} M) and in thin film state (Table 2, Fig. 4a-b). The UV–Vis spectra of **P1** and **P3** closely resembled those of the polymers synthesized using alternative methods [43,57], suggesting similar photophysical properties. Polymers **P2** and **P3** exhibited bandgaps (estimated from the absorption onset of UV–vis spectra in the thin film state) of 1.89 eV and 1.76 eV, respectively. Notably, the E_g^{opt} of **P1**, which consists of BTz and DPP units, decreased significantly to 1.29 eV, resulting in near-infrared absorption. In addition, the random terpolymers **P4** and **P5** displayed broad UV–vis spectra with a half maximum absorption range from 350 to 1000 nm, enabling absorption of visible and near-infrared light. This unique absorption property originates from the combined contribution of three different components, suggesting that **P4** and **P5** could be promising materials for high-performance light-harvesting applications [15,60]. It is important to note that the band gap correlation of the polymers, as calculated from the UV–Vis results, aligns with the DFT calculation results (Table S9), confirming the successful synthesis of the polymers using the aerobic DArP method with minimal structural errors. Overall, the BTz-based polymers exhibited similar E_g^{opt} values to the π-CPs synthesized by other methods, highlighting the versatility of the current DArP system and its potential to generate useful π-CPs for various applications in polymer electronics.

Next, we conducted fluorescence measurements of these polymers in a chloroform solution state (1.0×10^{-6} M) (Fig. 4c-d). The emission bands appeared in different wavelength regions depending on the

Table 2
Optical and thermal characteristics of **P1-P5**.

Polymer	$CHCl_3$ solution			Thin film			T_m (°C)[c]
	$\lambda_{abs,max}$ (nm)	$\lambda_{FL,max}$ (nm)	Δ (nm)[a]	$\lambda_{abs,max}$ (nm)	$\lambda_{abs,onset}$ (nm)	E_g^{opt} (eV)[b]	
P1	739	–	–	745	964	1.29	168
P2	509	577	68	551	657	1.89	n.d
P3	550	642	92	574	705	1.76	n.d
P4	716	–	–	730	905	1.37	n.d
P5	727	–	–	745	938	1.32	n.d

[a] Stokes shift in solution, $\Delta = \lambda_{FL,max} - \lambda_{abs,max}$. [b] $E_g^{opt} = 1240/\lambda_{abs,onset}$. [c] Determined by second heating DSC traces with heating rate of 10 °C/min under nitrogen. n.d: No obvious thermal transition was observed from 25 to 300 °C.

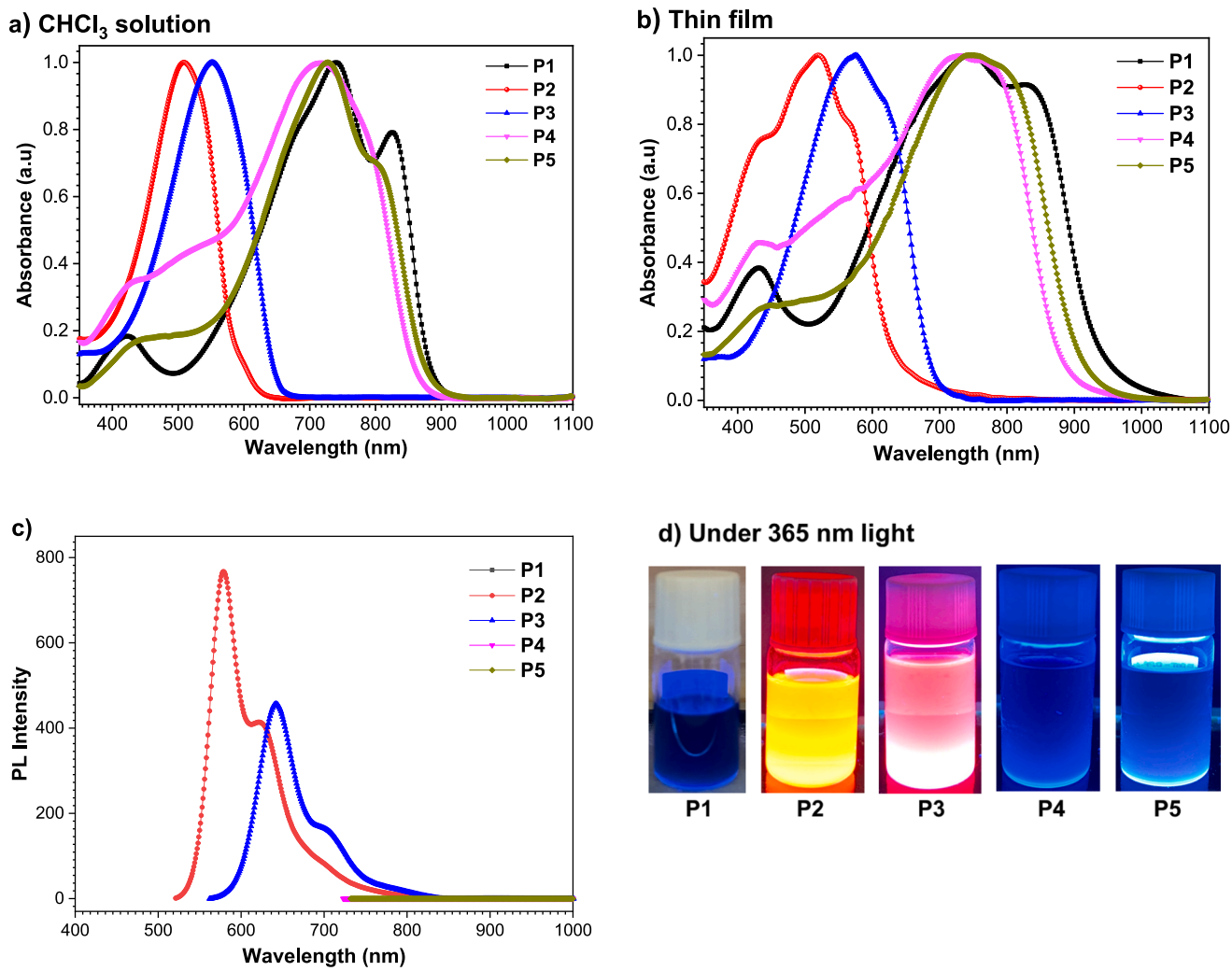


Fig. 4. Normalized UV–Vis absorption spectra of **P1–P5**: **a**) in chloroform solution and **b**) in thin solid films; **c**) Emission spectra of **P1–P5** in chloroform (1 μM) with excited at $\lambda_{abs,\ max}$; **d**) Photographs of polymers in chloroform taken under 365 nm light.

comonomers of the benzotriazole-based polymers. Polymers **P1**, **P4**, and **P5**, which contained DPP units, showed minimal emission in the 400–1000 nm range. On the other hand, **P2** and **P3** exhibited strong fluorescence with emission peaks at 577 nm and 642 nm with PL intensities of 780 and 468, respectively. These findings indicate that the BTz-DPP block acts as an acceptor component in terpolymers **P4** and **P5**, resulting in complete intramolecular fluorescence quenching [61]. This suggests its potential usefulness in organic solar cells.

Thermal properties. The thermal properties of polymers **P1-P5** were analyzed by the TGA and DSC methods under a nitrogen atmosphere with a heating rate of 10 °C/min, and the results are summarized in Table 2. All polymers exhibited good thermal stability with the onset point of decomposition temperatures (T_d) at 5 % weight loss over 350 °C, which is suitable for various organic electronic applications (As shown in Figure S19). The DSC thermograms of **P2-P5** showed no significant thermal transition between 25 and 300 °C, whereas **P1** exhibited a melting peak (T_m) at 168 °C (Figure S20).

3. Conclusions

In summary, we have developed a rapid method for synthesizing a variety of D–A-type benzotriazole-based π-CPs with high molecular masses (M_n > 20 kDa) using a Pd-catalyzed direct arylation polycondensation under aerobic conditions. This method is simple to carry out, environmentally friendly, and cost-effective. It allows for the production of various high-performance π-CPs with wide absorption ranges and complex microstructures, including D–A-type and A–A-type alternating copolymers, as well as A–A–A-type and A–A–D-type random terpolymers. Overall, these findings provide a versatile polycondensation tool for creating semiconducting polymers that are widely used in state-of-the-art plastic electronic applications. Our future research will explore the applicability of this aerobic DArP synthetic protocol for generating a variety of π-conjugated polymers and access their suitability to be used as materials for organic field-effect transistors and organic solar cells.

4. Experimental section

Detailed synthesis and characterizations of all compounds, and details of equipment are given in Supporting Information.

CRediT authorship contribution statement

Tam Huu Nguyen: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Hai Le Tran:** Validation, Software, Data curation. **Hai Minh Phan:** Validation, Software, Data curation. **Minh Duy Hoang:** Validation, Software, Data curation. **Le-Thu T. Nguyen:** Writing – review & editing, Formal analysis, Data curation. **Tin Chanh Duc Doan:** Visualization, Investigation. **Mai Ha Hoang:** Writing – review & editing, Formal analysis, Data

curation. **Quoc-Thiet Nguyen:** . **Ha Tran Nguyen:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113369.

References

[1] W. Hu, F. Bai, X. Gong, X. Zhan, H. Fu, T. Bjornholm, Organic optoelectronics, John Wiley & Sons, 2012.
[2] M. Geoghegan, G. Hadziioannou, Polymer electronics, Oxford Master Physics (2013).
[3] Z.B. Henson, K. Müllen, G.C. Bazan, Design strategies for organic semiconductors beyond the molecular formula, Nat. Chem. 4 (9) (2012) 699–704.
[4] Z. Peng, N. Stingelin, H. Ade, J.J. Michels, A materials physics perspective on structure–processing–function relations in blends of organic semiconductors, Nat. Rev. Mater. 8 (7) (2023) 439–455.
[5] C. Kunkel, J.T. Margraf, K. Chen, H. Oberhofer, K. Reuter, Active discovery of organic semiconductors, Nat. Commun. 12 (1) (2021) 2422.
[6] S.E. Root, S. Savagatrup, A.D. Printz, D. Rodriquez, D.J. Lipomi, Mechanical properties of organic semiconductors for stretchable, highly flexible, and mechanically robust electronics, Chem. Rev. 117 (9) (2017) 6467–6499.
[7] G. Zhang, F.R. Lin, F. Qi, T. Heumüller, A. Distler, H.-J. Egelhaaf, N. Li, P.C. Y. Chow, C.J. Brabec, A.K.Y. Jen, H.-L. Yip, Renewed Prospects for Organic Photovoltaics, Chem. Rev. 122 (18) (2022) 14180–14274.
[8] G.-J.-N. Wang, A. Gasperini, Z. Bao, Stretchable polymer semiconductors for plastic electronics, Adv. Electron. Mater. 4 (2) (2018) 1700429.
[9] A.J. Heeger, Semiconducting polymers: the third generation, ChSRv 39 (7) (2010) 2354–2371.
[10] R.M. Pankow, B.C. Thompson, The development of conjugated polymers as the cornerstone of organic electronics, Poly 207 (2020) 122874.
[11] A. Armin, W. Li, O.J. Sandberg, Z. Xiao, L. Ding, J. Nelson, D. Neher, K. Vandewal, S. Shoaee, T. Wang, H. Ade, T. Heumüller, C. Brabec, P. Meredith, A history and perspective of non-fullerene electron acceptors for organic solar cells, Adv. Energy Mater. 11 (15) (2021) 2003570.
[12] K. Müllen, W. Pisula, Donor-acceptor polymers, J. Am. Chem. Soc. 137 (30) (2015) 9503–9505.
[13] G. Yafei, Z. Kai, T. Yijie, Z. Chaoyang, C. Haifeng, Synthesis and electrochromic properties of soluble 4,9-dihydro-s-indaceno[1,2-b:5,6-b']dithiophene-embedded conjugated polymers and their applications on preparing green-to-transmissive electrochromics via solution co-processing, Sol. Energy Mater. Sol. Cells 195 (2019) 182–190.
[14] C. Yan, S. Barlow, Z. Wang, H. Yan, A.K.Y. Jen, S.R. Marder, X. Zhan, Non-fullerene acceptors for organic solar cells, Nat. Rev. Mater. 3 (3) (2018) 18003.
[15] D. Dang, D. Yu, E. Wang, Conjugated donor-acceptor terpolymers toward high-efficiency polymer solar cells, Adv. Mater. 31 (22) (2019) 1807019.
[16] T.R. Pavase, H. Lin, Q.-u.-a. Shaikh, S. Hussain, Z. Li, I. Ahmed, L. Lv, L. Sun, S.B.H. Shah, M.T. Kalhoro, Recent advances of conjugated polymer (CP) nanocomposite-based chemical sensors and their applications in food spoilage detection: a comprehensive review, Sens. Actuat. B: Chem. 273 (2018) 1113–1138.
[17] S.-J. Zou, Y. Shen, F.-M. Xie, J.-D. Chen, Y.-Q. Li, J.-X. Tang, Recent advances in organic light-emitting diodes: toward smart lighting and displays, Mater. Chem. Front. 4 (3) (2020) 788–820.
[18] Z. Zhang, G. Chen, Y. Xue, Q. Duan, X. Liang, T. Lin, Z. Wu, Y. Tan, Q. Zhao, W. Zheng, L. Wang, F. Wang, X. Luo, J. Xu, J. Liu, B. Lu, Fatigue-resistant conducting polymer hydrogels as strain sensor for underwater robotics, Adv. Funct. Mater. 33 (42) (2023) 2305705.
[19] Z. Wu, Q. Zhao, X. Luo, H. Ma, W. Zheng, J. Yu, Z. Zhang, K. Zhang, K. Qu, R. Yang, N. Jian, J. Hou, X. Liu, J. Xu, B. Lu, Low-cost fabrication of high-performance fluorinated polythiophene-based Vis–NIR electrochromic devices toward deformable display and camouflage, Chem. Mater. 34 (22) (2022) 9923–9933.
[20] W. Zhou, J. Xu, Progress in conjugated polyindoles: synthesis, polymerization mechanisms, properties, and applications, Polym. Rev. 57 (2) (2017) 248–275.
[21] X. Zhou, S. Pang, B. Wu, J. Zhou, H. Tang, K. Lin, Z. Xie, C. Duan, F. Huang, Y. Cao, Noncovalent interactions induced by fluorination of the central core improve the photovoltaic performance of A-D-A′-D-A-type nonfused ring acceptors, ACS Appl. Energy Mater. 5 (6) (2022) 7710–7718.
[22] L. Ding, Z.-D. Yu, X.-Y. Wang, Z.-F. Yao, Y. Lu, C.-Y. Yang, J.-Y. Wang, J. Pei, Polymer semiconductors: synthesis, processing, and applications, Chem. Rev. 123 (12) (2023) 7421–7497.
[23] B. Carsten, F. He, H.J. Son, T. Xu, L. Yu, Stille polycondensation for synthesis of functional materials, Chem. Rev. 111 (3) (2011) 1493–1528.
[24] T. Yokozawa, Y. Ohta, Transformation of step-growth polymerization into living chain-growth polymerization, Chem. Rev. 116 (4) (2016) 1950–1968.
[25] J.-R. Pouliot, F. Grenier, J.T. Blaskovits, S. Beaupré, M. Leclerc, Direct (hetero) arylation polymerization: simplicity for conjugated polymer synthesis, Chem. Rev. 116 (22) (2016) 14225–14274.
[26] N.S. Gobalasingham, B.C. Thompson, Direct arylation polymerization: a guide to optimal conditions for effective conjugated polymers, Prog. Polym. Sci. 83 (2018) 135–201.
[27] M. Wakioka, F. Ozawa, Highly efficient catalysts for direct arylation polymerization (DArP), Asian J. Org. Chem. 7 (7) (2018) 1206–1216.
[28] T. Bura, J.T. Blaskovits, M. Leclerc, Direct (hetero)arylation polymerization: trends and perspectives, J. Am. Chem. Soc. 138 (32) (2016) 10056–10071.
[29] S. Kowalski, S. Allard, K. Zilberberg, T. Riedl, U. Scherf, Direct arylation polycondensation as simplified alternative for the synthesis of conjugated (co) polymers, Prog. Polym. Sci. 38 (12) (2013) 1805–1814.
[30] C. Wu, H. Chen, K. Lin, J. Li, Y. Wang, B. Lu, Recent advances in electrochromic conjugated polymers prepared by direct (hetero) arylation polymerization, SynMe 306 (2024) 117632.
[31] C. Wu, H. Chen, J. Tan, D. Zhou, H. Liang, S. Zhen, H. Liu, Y. Wang, K. Lin, Electrochromic conjugated polymers containing benzotriazole and thiophene performing sub-second response time and 916 cm2 C−1 superb coloration efficiency, Sol. Energy Mater. Sol. Cells 257 (2023) 112355.
[32] A.S. Dudnik, T.J. Aldrich, N.D. Eastham, R.P.H. Chang, A. Facchetti, T.J. Marks, Tin-free direct C-H arylation polymerization for high photovoltaic efficiency conjugated copolymers, J. Am. Chem. Soc. 138 (48) (2016) 15699–15709.
[33] R. Matsidik, H. Komber, A. Luzio, M. Caironi, M. Sommer, Defect-free naphthalene diimide bithiophene copolymers with controlled molar mass and high performance via direct arylation polycondensation, J. Am. Chem. Soc. 137 (20) (2015) 6705–6711.
[34] M. Leclerc, S. Brassard, S. Beaupré, Direct (hetero)arylation polymerization: toward defect-free conjugated polymers, Polym. J. 52 (1) (2020) 13–20.
[35] Q. Wang, R. Takita, Y. Kikuzaki, F. Ozawa, Palladium-catalyzed dehydrohalogenative polycondensation of 2-bromo-3-hexylthiophene: an efficient approach to head-to-tail poly(3-hexylthiophene), J. Am. Chem. Soc. 132 (33) (2010) 11420–11421.
[36] Y. Li, X. Yuan, S. Kim, Y. Zhang, D. Xie, X. Tan, C. Yang, X. Huang, F. Huang, Y. Cao, C. Duan, Revealing the molecular weight effect on highly efficient polythiophene solar cells, ACS Appl. Mater. Interfaces 15 (24) (2023) 29341–29351.
[37] H. Chen, K. Lin, H. Liang, J. Tan, D. Zhou, X. Zhang, F. Liu, Y. Wang, Molecular weight and linking position (2,7 vs 3,6) effects on electrochromic properties of alternating carbazole-EDOT copolymers obtained by direct (hetero) arylation polymerization, SynMe 291 (2022) 117200.
[38] F. Grenier, K. Goudreau, M. Leclerc, Robust direct (hetero)arylation polymerization in biphasic conditions, J. Am. Chem. Soc. 139 (7) (2017) 2816–2824.
[39] A. Ichige, H. Saito, J. Kuwabara, T. Yasuda, J.-C. Choi, T. Kanbara, Facile synthesis of thienopyrroledione-based π-conjugated polymers via direct arylation polycondensation under aerobic conditions, Macromolecules 51 (17) (2018) 6782–6788.
[40] X. Chen, A. Ichige, J. Chen, I. Fukushima, J. Kuwabara, T. Kanbara, Facile access to conjugated polymers under aerobic conditions via Pd-catalyzed direct arylation and aryl amination polycondensation, Poly 207 (2020) 122927.
[41] C. Carlucci, D. Conelli, O. Hassan Omar, N. Margiotta, R. Grisorio, G.P. Suranna, Toward eco-friendly protocols: insights into direct arylation polymerizations under aerobic conditions in anisole, Polym. Chem. 14 (3) (2023) 343–351.
[42] Q. Liu, S.E. Bottle, P. Sonar, Developments of diketopyrrolopyrrole-dye-based organic semiconductors for a wide range of applications in electronics, Adv. Mater. 32 (4) (2020) 1903882.
[43] M. Gruber, S.-H. Jung, S. Schott, D. Venkateshvaran, A.J. Kronemeijer, J. W. Andreasen, C.R. McNeill, W.W.H. Wong, M. Shahid, M. Heeney, J.-K. Lee, H. Sirringhaus, Enabling high-mobility, ambipolar charge-transport in a DPP-benzotriazole copolymer by side-chain engineering, Chem. Sci. 6 (12) (2015) 6949–6960.
[44] S.J. Firsan, V. Sivakumar, T.J. Colacot, Emerging trends in cross-coupling: twelve-electron-based L1Pd(0) catalysts, their mechanism of action, and selected applications, Chem. Rev. 122 (23) (2022) 16983–17027.
[45] M. Wakioka, N. Ichihara, Y. Kitano, F. Ozawa, A highly efficient catalyst for the synthesis of alternating copolymers with thieno[3,4-c]pyrrole-4,6-dione units via direct arylation polymerization, Macromolecules 47 (2) (2014) 626–631.
[46] W. Gao, Z. He, Y. Qian, J. Zhao, Y. Huang, General palladium-catalyzed aerobic dehydrogenation to generate double bonds, Chem. Sci. 3 (3) (2012) 883–886.

[47] T. Yurino, S. Saito, M. Ichihashi, T. Ohkuma, Aerobic allylic amination catalyzed by a Pd(OAc)2/P(OPh)3 system with low catalyst loading, J. Org. Chem. 87 (5) (2022) 2864–2872.
[48] M. Wakioka, Y. Nakamura, M. Montgomery, F. Ozawa, Remarkable ligand Effect of P(2-MeOC6H4)3 on palladium-catalyzed direct arylation, Organometallics 34 (1) (2015) 198–205.
[49] X. Wang, M. Wang, Synthesis of donor–acceptor conjugated polymers based on benzo[1,2-b:4,5-b′]dithiophene and 2,1,3-benzothiadiazole via direct arylation polycondensation: towards efficient C-H activation in nonpolar solvents, Polym. Chem. 5 (19) (2014) 5784–5792.
[50] H.-Y. Sun, S.I. Gorelsky, D.R. Stuart, L.-C. Campeau, K. Fagnou, Mechanistic analysis of azine N-oxide direct arylation: evidence for a critical role of acetate in the Pd(OAc)2 precatalyst, J. Org. Chem. 75 (23) (2010) 8180–8189.
[51] J. Blaskovits, M. Leclerc, Direct (Hetero)Arylation Polymerization for the Preparation of Conjugated Polymers, 2019, pp. 195-238.
[52] C.H.T. Nguyen, T.H. Nguyen, T.P.L. Nguyen, H. Le Tran, T.H. Luu, C.D. Tran, Q.-T. Nguyen, L.-T.-T. Nguyen, T. Yokozawa, H.T. Nguyen, Aerobic direct arylation polycondensation of N-perylenyl phenoxazine-based fluorescent conjugated polymers for highly sensitive and selective TNT explosives detection, Dyes Pigm. 219 (2023) 111613.
[53] M. Yasa, L. Toppare, Thieno[3,4-c]pyrrole-4,6-dione-based conjugated polymers for nonfullerene organic solar cells, Macromol. Chem. Phys. 223 (5) (2022) 2100421.
[54] C. Zhao, F. Yang, D. Xia, Z. Zhang, Y. Zhang, N. Yan, S. You, W. Li, Thieno[3,4-c]pyrrole-4,6-dione-based conjugated polymers for organic solar cells, ChCom 56 (72) (2020) 10394–10408.
[55] L.V. Kayser, D.J. Lipomi, Stretchable conductive polymers and composites based on PEDOT and PEDOT:PSS, Adv. Mater. 31 (10) (2019) 1806133.
[56] J. Kuwabara, K. Yamazaki, T. Yamagata, W. Tsuchida, T. Kanbara, The effect of a solvent on direct arylation polycondensation of substituted thiophenes, Polym. Chem. 6 (6) (2015) 891–895.
[57] H. Kim, J. Lee, T. Kim, M. Cho, T.-L. Choi, Precision synthesis of various low-bandgap donor-acceptor alternating conjugated polymers via living Suzuki-Miyaura catalyst-transfer polymerization, Angew. Chem. Int. Ed. 61 (31) (2022) e202205828.
[58] K. Lin, H. Chen, H. Liang, J. Tan, D. Zhou, X. Zhang, F. Liu, Y. Wang, Benzotriazole-EDOT electrochromic conjugated polymers show sub-second response time and 774 cm2 C−1 coloration efficiency, Njch 46 (35) (2022) 16684–16692.
[59] X. Yang, Y. Shao, S. Wang, M. Chen, B. Xiao, R. Sun, J. Min, Processability Considerations for Next-Generation Organic Photovoltaic Materials, Adv. Mater. n/a(n/a) 2307863.
[60] H. Luo, Z. Liu, D. Zhang, Conjugated D-A terpolymers for organic field-effect transistors and solar cells, Polym. J. 50 (1) (2018) 21–31.
[61] B.R. Gautam, R. Younts, J. Carpenter, H. Ade, K. Gundogdu, The role of FRET in non-fullerene organic solar cells: implications for molecular design, Chem. A Eur. J. 122 (15) (2018) 3764–3771.